Rafay Khan

Principal AI Engineer

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Principal AI/ML Engineer with 8+ years building and scaling production systems across LLMs, computer vision, recommendation systems, and edge inference. Leading technical delivery across AI, backend, and infrastructure teams, owning systems from model design and evaluation through deployment, monitoring, and incident response. Delivered multilingual assistants handling 50,000+ daily conversations, real-time computer vision across 1,500+ smart carts, and recommendation models trained on 2M+ user interactions.

PROFESSIONAL EXPERIENCE

VisionX – *AI development and consulting firm building custom ML solutions*

Principal AI Engineer | May 2024 – Present *Dubai, UAE / Islamabad, Pakistan*

- Lead technical delivery across AI, backend, and infrastructure teams for **a multilingual LLM-powered support assistant** used **by Iraq's largest payments processor**, handling **50,000+ customer conversations per day** across **Arabic, Kurdish, and English** chat and voice on a platform **serving 30M+ users**.
- Kept **end-to-end response times near 5s** under load by deploying the service on **AWS ECS** with **Terraform**-managed autoscaling, load-tested at **2,500+ requests per second**.
- Enabled customers to receive support through text, images, and real-time voice by building a multimodal inference stack **with FastAPI, OpenAI models, and** self-hosted models through **SGLang**.
- Reduced customer and operational risk by adding **guardrails**, **fallback rules**, and **human escalation** for low-confidence answers, unsafe requests, and tool or API failures.
- Made prompt and model releases safer by building **offline evaluation suites** that caught pre-deployment regressions in multilingual response quality, guardrail compliance, human escalation, and resolution quality, with production monitoring tracking resolution quality and escalation patterns after release.
- Improved reliability across engineering teams by defining **OpenAPI** contracts and instrumenting end-to-end chat and voice flows with **OpenTelemetry**, making integration failures and production incidents easier to diagnose.

Lightly.ai – *YC-backed data curation platform; Lightly Edge selects high-value data on-device*

Senior Machine Learning Engineer (Consultant) | January 2024 – April 2024 *Zürich, Switzerland*

- Built Rust and ONNX Runtime inference components for **LightlyEdge**, a sub-200MB on-device **smart data selection SDK** that identifies high-signal samples in real-time on **NVIDIA Jetson** hardware, reducing data transfer and storage across edge ML pipelines.
- Improved production reliability by standardizing preprocessing, output handling, batching, and component interfaces across the SDK's model-serving workflow.
- Lowered on-device compute overhead while preserving predictable selection results by applying **model quantization** and **pruning** for resource-constrained deployments.

Veeve.io – *Venture-backed smart-cart technology using computer vision and sensor fusion; raised $6.7M during tenure*

Senior Machine Learning Engineer – Team Lead | January 2020 – November 2023 *Seattle, USA*

- Led ML team in development of **real-time in-cart video intelligence system,** enabling smart shopping-carts to recognize shopper actions and track products, running at **20 FPS** across production retail environments.
- Kept recognition quality and real-time performance reliable in noisy stores by owning the **full model lifecycle, including evaluation, deployment, monitoring, and annotation workflows**, while balancing **precision, recall, latency**, and **edge-device constraints**.
- Supported international rollout, including work with Albertsons and Kroger, by scaling inference to **1,500+ simultaneously operating carts** across the **US, Europe, and the Middle East**.
- Maintained **p95 latencies of 20 ms for detection and 30 ms for classification** while serving custom computer-vision models on **GCP** and **Kubernetes** with **NVIDIA Triton**.
- Reduced release risk and accelerated model iteration by building a **GitOps deployment pipeline** with controlled promotion and rollback of model versions.
- Cut debugging time and improved reproducibility by tracking training runs, datasets, and model artifacts in **Weights & Biases**.

▶ Watch 2-minute early prototype demo of real-time in-cart activity detection

(In this video, the model identifies seven fine-grained actions including insert, remove, rummage, and theft-like behavior. Visualized via a step-function and live confidence bar chart. Since the demo, I expanded class coverage, especially for theft detection, and optimized accuracy for both edge and cloud deployment.)

Presidential Initiative for Artificial Intelligence and Computing (PIAIC)

Assistant Instructor** (part-time) **| September 2019 – September 2020 *Islamabad, Pakistan*

- Helped learners translate AI theory into working assignments by designing coursework and reviewing technical work as an assistant instructor after ranking in **the top 1.7% of the program's early cohort**.

SPECIAL PROJECT

Skinsight.me – *Personalized AI-driven skincare product recommendation platform*

Co-Founder *Dubai, UAE*

- Built an AI skincare recommendation product end-to-end, spanning **frontend in Next.js**, **backend in FastAPI**, **Temporal-managed durable data ingestion pipelines** for continuous catalog growth, and an **agentic product discovery assistant** that recommends skincare products through conversational flows; all **scaled on AWS ECS**.
- Trained a **neural collaborative filtering system** on **60K+ products** and **2M+ user data points** to generate personalized recommendations and high-intent alternatives.
- Designed architecture for scalable iteration: modular pipelines, reproducible training runs, and clear interfaces between data acquisition, modeling, and product delivery.

RESEARCH EXPERIENCE

Mila

Research Fellow *(part-time)* ***| April 2022 – June 2024*** *Quebec, Canada (Remote)*

Research Fellow as part of the Fatima Fellowship. The primary investigator on the project ***"More than more data? Comparing training with data augmentation vs IID data"***.

Supervisor: Alex Hernandez-Garcia

- Designed controlled experiments comparing augmented and IID training across five synthetic task families to isolate their effects on model robustness and convergence.
- Built a modular **PyTorch Lightning** and Hydra workbench for reproducible single-run and multi-run experiments, including automated sample-size sweeps from 20 to 10,000. (https://github.com/alexhernandezgarcia/data-augmentation).
- Isolated the effects of individual transformations by developing controlled augmentation methods and synthetic datasets that reduced confounding between data quantity and augmentation quality.

TUKL-NUST R&D Lab

Research Assistant | September 2016 – June 2017 *Islamabad, Pakistan*

Vision research team lead for the project ***"Fish Biodiversity Estimation by Low-Cost Non-Destructive Video Based Sampling (FIBEVID)."*** This project was in collaboration with Pakistan Fisheries Department, University of Kaiserslautern (TUKL) to help create a non-destructive fish bio-diversity estimation tool for Pakistan's National fish – the Mahasher. Project went on to receive funding from Hochschule RheinMain and DAAD (German Academic Exchange Service).

Supervisors: Dr. Faisal Shafiat, Dr. Imran Malik

(https://www.hs-rm.de/de/fachbereiche/design-informatik-medien/forschungsprofil/fibevid/)

- Enabled low-cost, non-destructive monitoring of Pakistan's Mahasher fish **by creating and publishing the first 100,000-image labeled dataset** captured in the species' natural river habitat with Fisheries Department and German university partners.

Research Assistant | June'16 – Sep'16 *(part-time)*

Summer intern at the TUKL-NUST R&D lab. Worked in collaboration with researchers at University of Kaiserslautern detect forged bank cheques under the project "***Forgery detection and segmentation of handwritten text through Hyperspectral Image Analysis***."

Supervisor: Dr. Imran Malik

- Supported non-destructive detection of altered bank cheques by statistically analyzing hyperspectral images to distinguish different ink types in a TUKL-NUST research collaboration.

EDUCATION

National University of Sciences and Technology (NUST) *Islamabad, Pakistan*

Bachelor of Science in Computer Science (BSCS) ***2013 – 2017***

Rector's High Achiever's List 2017 Final Year Project - Fish Biodiversity Estimation by Low-Cost Non-Destructive Video Based Sampling, FIBEVID

SKILLS

AI/ML: Applied LLMs, agentic systems, OpenAI APIs, evaluation and guardrails, computer vision, recommendation systems, PyTorch, TensorFlow

Software Engineering: Python, C/C++, Rust, Java, JavaScript/TypeScript, FastAPI, Next.js

Cloud and production: AWS, GCP, Docker, Kubernetes, Terraform, GitOps, SGLang, NVIDIA Triton, ONNX Runtime, Weights & Biases, OpenTelemetry

Languages: Urdu (native), English (fluent), Spanish (basic), German (basic)

SELECT HONORS & AWARDS

- **Fatima Fellowship, 2022**. Selected in top 53 fellows from an applicant pool of over 700 from 68 different countries
- **Rector's High Achievers List, 2017** final year project – FIBEVID
- **Research grant DAAD (German Academic Exchange Service), 2017** final year project – FIBEVID
- **3rd best project SEECS Open House**, **2017**. Department-wide project competition at NUST – FIBEVID
- **2nd best project at COMPPEC, EME college**, **2017**. Nation-wide software project competition.

SELECT BLOGS

- **Nothing but NumPy: Understanding & Creating Neural Networks with Computational Graphs from Scratch**
  - 50K+ views; KDnuggets Gold Award
- **Nothing but NumPy: Understanding & Creating Binary Classification Neural Networks with Computational Graphs from Scratch**
  - 48K+ views; Published in Towards Data Science
- **Where did the Binary Cross-Entropy Loss Function come from?**
- **Why Using Mean Squared Error (MSE) Cost Function for Binary Classification is a Bad Idea?**
- **How do TensorFlow and Keras implement Binary Classification and the Binary Cross-Entropy function?**